

Compose, Don't Tune?

A Retrospective Fixed-Panel Study of Base-Adapter Score Composition for Safety Guards

Reza Rahimi

JazzX AI, Los Altos, CA, USA

reza.rahimi@jazzx.ai

Working draft — July 2026

Study status. This manuscript is a research draft built from *retrospective* clean-v2 evidence. The transfer benchmarks were withheld from SFT but inspected during method development. No separately locked prospective Paper B study has run. Sections describing future validation are protocol, not results.

Abstract

Fine-tuning a compact language model into a binary safety guard can improve discrimination on benchmark sources represented during adaptation while weakening transfer to other datasets. We ask whether keeping the same checkpoint's pre-adaptation decision in an output-space composition can recover transfer relative to its SFT adapter. Using the lock-verified per-row scores from a companion study, we retrospectively compare 4 unadapted instruction checkpoints, 5 LoRA-SFT adapters per checkpoint, and a fixed average of separately calibrated unsafe probabilities. Relative to SFT, the observed composition contrast is -0.019 in represented-source macro average precision (AP), with a descriptive paired-bootstrap 95% interval [-0.031, -0.010], and +0.076 on dataset-held-out transfer ([+0.058, +0.093]). The transfer contrast relative to the base is heterogeneous: positive for 2 checkpoints, near zero for one, and negative for Qwen3-4B. Thresholds chosen for a 5% calibration false-positive-rate (FPR) target yield 11.4% transfer macro-FPR for the composition, so rank recovery does not imply calibration transfer. The method also requires two inference passes. Because the operator and transfer results informed this analysis, these are conditional fixed-panel retrospective estimates, not confirmatory evidence. They motivate a separately locked prospective comparison against WiSE-FT, an equal-inference-cost SFT+SFT ensemble, and matched-compute preservation tuning.

1 Introduction

Parameter-efficient adaptation makes it inexpensive to specialize an instruction model into a prompt-safety guard. LoRA, for example, updates a low-rank parameterization rather than all model weights [1]. Yet specialization creates a practical tension: a guard can become better at the policies and benchmark sources represented during SFT while losing useful ranking information on dataset-held-out sources. The companion Paper A study measures this tension under a controlled base-to-adapter design [3]. This draft asks the next, narrower question: can a fixed composition recover some transfer relative to SFT without another training run?

The candidate rule is deliberately simple. For one input, run the unadapted checkpoint and one of its SFT adapters, calibrate their unsafe scores using development calibration rows, and average the two calibrated probabilities with fixed equal weights. Here *base* means unadapted to the guard SFT recipe; it does not mean pretrained from scratch, unaligned, or generally untuned. The rule is output-space and therefore does not require the two systems to expose interpolable weights, but it does require aligned labels, compatible unsafe-score semantics, separate calibration, and two inference passes.

The current evidence is a feasibility result, not the final Paper B study. Paper A’s score table and lock establish what inputs were analyzed, but they do not prospectively bind a composition operator invented later. Moreover, some transfer benchmarks and earlier results were seen during development. We therefore use the current panel to estimate a retrospective retention–recovery trade-off, expose failure modes, and write a protocol that can later be frozen before a genuinely uninspected cohort is evaluated.

This draft makes three bounded contributions:

1. a generated, provenance-gated summary of the clean-v2 retrospective composition pilot;
2. an explicit claim boundary separating score composition from prior weight-space and calibrated-ensemble methods; and
3. an executable prospective-protocol contract that fails claim-bearing validation until the cohort, margin, controls, statistics, systems measurements, and software lock are fixed.

It does *not* claim Pareto dominance, universal improvement, a causal mechanism, a validated base-competence rule, or deployment readiness.

2 Related work and novelty boundary

The underlying ensemble idea is not new. Calibrated ensembling has been shown to mitigate in-distribution and distribution-shift trade-offs in image classification when component models are calibrated using in-distribution data [2]. WiSE-FT interpolates the weights of a zero-shot model and its fine-tuned descendant to improve robustness under shift [5]. Model soups average compatible fine-tuned weights and can avoid the runtime cost of a conventional multi-pass ensemble [4].

This study does not present score averaging as a novel primitive. Its narrower target is a safety-guard setting in which a single guard SFT may improve represented-source ranking while degrading dataset-held-out ranking. Output-space composition is architecture-agnostic in the limited sense that it needs comparable decision scores rather than interpolable parameters. That portability comes with a systems cost: both components must run. A submission-grade study must therefore compare against actual WiSE-FT rescoring, not merely discuss it, and must measure whether an SFT+SFT two-pass ensemble obtains the same benefit. Without the latter control, any gain could be generic variance reduction from a second model pass rather than information uniquely retained by the base.

3 Evidence status and research questions

Table 1 separates the evidence tiers used in this draft. The distinction is substantive: clean execution provenance does not erase prior exposure to evaluation sources, and a Paper A lock cannot bind Paper B code written after it.

Table 1: Evidence tiers. Only the first two rows exist; the prospective row is planned.

Tier	What it establishes	Current status
Paper A clean-v2 scores	Row-keyed base and SFT scores bound to the Paper A execution contract	Complete
Retrospective composition	Reproducible post-lock analysis of fixed, previously exposed transfer benchmarks	<code>clean_v2_retrospective_estimation</code>
Prospective Paper B	Frozen operator, margins, controls, model panel, uninspected cohort, and analysis	Not executed

The pilot addresses three research questions, all conditional on the fixed 4-checkpoint panel:

RQ1: represented retention.

What is the composition-minus-SFT change on sources represented during SFT? The pilot estimates the loss; it does not label that loss noninferior because no margin was prospectively justified.

RQ2: transfer recovery.

What is the composition-minus-SFT change on dataset-held-out transfer benchmarks, and how does the composition compare descriptively with the base?

RQ3: heterogeneity hypothesis.

Can base competence measured on a locked development cohort predict composition-minus-SFT recovery on a disjoint prospective cohort? The current 4 points motivate this question but cannot validate a decision rule.

4 Retrospective pilot methods

4.1 Panel, data roles, and score artifact

The pilot reuses Paper A’s fixed panel: Qwen2.5-1.5B, SmolLM2-1.7B, SmolLM3-3B, and Qwen3-4B, with 5 LoRA-SFT seeds per checkpoint. Evaluation sources are grouped into represented-source tests and dataset-held-out transfer tests. Dataset-held-out means those rows were not used for SFT; it does not mean the benchmark identities or earlier outcomes were unseen by the researcher. All absolute and contrast values consumed by this manuscript are generated from the canonical composition JSON rather than copied into the TeX source.

The publication adapter refuses legacy evidence, a nonfinal run, a changed operator, an altered seed panel, or a prospective flag. It binds the exact retrospective result with composition hash `92c2cbc3...`, Paper A lock `cabc8dee...`, and score artifact `b941ddba...`. These hashes make the draft stable; they do not constitute a prospective Paper B lock.

4.2 Fixed primary operator

Let $s_b(x)$ and $s_{a,j}(x)$ denote the unsafe scores for base b and SFT seed j . Let C_b and $C_{a,j}$ be calibrators fit only on the development calibration split. The fixed primary composition is

$$s_{\text{comp},j}(x) = \frac{1}{2}C_b(s_b(x)) + \frac{1}{2}C_{a,j}(s_{a,j}(x)). \quad (1)$$

The analyzer fixes equal weights without fitting reported test rows. Because the rule was formalized after earlier transfer exposure, this implementation fact is not human test blindness. The remaining score combiners are exposed ablations and cannot become primary after observing transfer.

4.3 Estimands and uncertainty

For regime $r \in \{\text{represented}, \text{transfer}\}$, the primary descriptive estimand is

$$\Delta_r = \text{macroAP}_r(\text{composition}) - \text{macroAP}_r(\text{SFT}). \quad (2)$$

AP is computed per benchmark with tie-aware ranking, macro-averaged over benchmarks, averaged over SFT seeds, and then averaged equally across the 4 fixed checkpoints. The composition- minus-base contrast is a secondary anchor. It is not the main success criterion because the goal is to recover transfer relative to SFT while retaining a bounded amount of represented-source performance.

Descriptive intervals use a paired family-by-seed hierarchical bootstrap. Family resampling is shared across both sides of a contrast; SFT seeds are resampled within checkpoint. The base has one scoring

pass, so seed resampling captures adapter-side variation but not uncertainty over base training seeds. The reported point beside each percentile interval is the observed contrast. A separate generated macro retains the bootstrap mean so the two quantities cannot be silently mixed.

The 4 checkpoints and 7 AP datasets are treated as a fixed panel, not random samples from model or benchmark populations. Intervals summarize row-family and adapter-seed variation conditional on this panel and protocol; they do not license universal model-family claims or formal population-level significance language.

4.4 Operating-point diagnostic

Macro-AP is threshold-free. A threshold is selected from development calibration negatives for a target FPR of 5%. The artifact retains both regimes; Table 4 reports transfer macro and pooled FPR/TPR. This secondary diagnostic is not a production guarantee.

5 Retrospective results

Table 2: Retrospective clean-v2 fixed-panel macro-AP. The calibrated average is the fixed primary operator. Logit averaging is an exposed ablation and cannot be promoted after observing transfer results.

Guard	Represented	Transfer	min(both)
Unadapted base	0.658	0.866	0.658
SFT adapter	0.982	0.807	0.807
Base+SFT calibrated average	0.962	0.883	0.883
Base+SFT logit average (ablation)	0.943	0.891	0.891

The calibrated composition changes represented-source macro-AP from 0.982 for SFT to 0.962, an observed difference of -0.019 with a paired-bootstrap percentile interval [-0.031, -0.010]. On transfer it changes macro-AP from 0.807 to 0.883, an observed difference of +0.076 [+0.058, +0.093]. These estimates describe a measured exchange: lower represented-source AP than SFT and higher transfer AP than SFT. Whether the represented loss is acceptable is a decision question requiring a prospectively supported noninferiority margin.

The panel-average composition-minus-base transfer contrast is +0.017 [+0.005, +0.030], but the panel mean hides important heterogeneity. Table 3 shows 2 positive observed composition-minus-base contrasts for SmolLM2 and Qwen2.5, an interval spanning zero for SmolLM3, and a negative contrast for Qwen3-4B. Thus the pilot supports transfer recovery relative to SFT on these checkpoints; it does not support a claim that composition beats both components per checkpoint.

Table 3: Transfer macro-AP by checkpoint. Deltas are observed contrasts with descriptive paired-bootstrap percentile intervals. Recovery relative to SFT is positive for every checkpoint, but comparison with base is heterogeneous.

Checkpoint	Base	SFT	Base+SFT	Δ vs. SFT [95% CI]	Δ vs. base [95% CI]
SmolLM2-1.7B	0.790	0.830	0.857	+0.027 [+0.013, +0.043]	+0.067 [+0.038, +0.095]
Qwen2.5-1.5B	0.819	0.780	0.855	+0.075 [+0.047, +0.103]	+0.036 [+0.008, +0.066]
SmolLM3-3B	0.910	0.823	0.907	+0.084 [+0.063, +0.104]	-0.003 [-0.012, +0.005]
Qwen3-4B	0.944	0.794	0.914	+0.120 [+0.082, +0.160]	-0.030 [-0.043, -0.018]

Logit averaging reaches transfer macro-AP 0.891, above the fixed primary in this exposed pilot. It remains an ablation. Selecting it after observing transfer would convert a descriptive comparison into an

unacknowledged tuning step. Likewise, the calibration-only convex search selects SFT weight 0.95; its development selection does not make it the precommitted primary operator.

Table 4: Realized transfer operating points after choosing thresholds for a 5% FPR target on calibration negatives. Rank recovery does not imply calibration transfer.

Guard	Macro TPR	Macro FPR	Pooled FPR
Unadapted base	0.517	0.081	0.043
SFT adapter	0.581	0.155	0.170
Base+SFT calibrated average	0.639	0.114	0.091

The operating-point result is less favorable than the ranking result. Against a 5% calibration target, realized transfer macro-FPR is 8.1% for the base, 15.5% for SFT, and 11.4% for the composition. Composition improves macro-TPR relative to both components in this diagnostic, but it still misses the FPR target. Any deployment claim based only on AP would omit this calibration-transfer failure.

6 What the pilot does not establish

Several tempting interpretations are not supported.

- **Not Pareto dominance.** Composition is lower than SFT on represented-source AP, and it is lower than base on transfer for Qwen3-4B. A supported noninferiority margin can define bounded retention; it cannot retroactively create literal Pareto improvement.
- **Not a base-competence law.** Regressing same-cohort base AP against composition-minus-base embeds the predictor in the outcome with a negative sign. That mathematical coupling can manufacture an association. The prospective protocol instead uses development-cohort competence to predict a composition-minus-SFT outcome on a disjoint cohort.
- **Not a mechanism.** Signal-destroying and within-class shuffle diagnostics can eliminate some explanations but cannot identify why composition changes ranking.
- **Not evidence that keeping the base is uniquely helpful.** The current pilot lacks the independently seeded SFT+SFT two-pass control. The effect may be generic ensembling.
- **Not free.** Output composition requires two inference passes and associated latency, throughput, memory, energy, and serving complexity. The pilot does not yet contain standardized systems measurements.

7 Prospective validation protocol — not yet executed

The repository contains a machine-validated draft contract, not a completed preregistration. A claim-bearing run must use a separate Paper B lock that binds the imported Paper A evidence, prospective cohort, source tree, analysis configuration, environment, RNGs, operator hierarchy, model panel, estimands, multiplicity policy, and systems protocol. The validator’s locked mode is designed to fail until every item below is concrete.

7.1 Cohort and blindness

The primary outcome cohort must be selected and content-hashed before inspection of its labels or model scores. It must be genuinely disjoint from the cohort used to estimate base competence. Calling a known public benchmark “held out” is insufficient if its earlier scores influenced operator development. Retrospective and prospective estimates will be reported separately rather than pooled.

7.2 Joint success rule

Before evaluation, the protocol must justify and freeze a represented-source noninferiority margin m . A prospective success rule should require both

$$\text{LCB}(\Delta_{\text{represented}}) > -m, \quad (3)$$

$$\text{LCB}(\Delta_{\text{transfer}}) > 0, \quad (4)$$

under pre-specified one-sided intervals and a multiplicity policy. The value of m cannot be chosen from the observed pilot loss. Even if both criteria pass, the claim should be bounded retention plus transfer recovery relative to SFT, not Pareto dominance.

7.3 Required controls

Table 5: Minimum prospective comparison set. “Missing” means no claim-bearing Paper B result exists yet, not that an implementation can never be built.

Arm	Purpose	Status
Base	Single-pass pre-guard-SFT anchor	Retrospective scores ready
SFT	Specialized single-pass reference	Retrospective scores ready
Base+SFT output	Fixed primary candidate in Equation (1)	Retrospective only
SFT+SFT output	Equal-inference-cost generic ensemble control; aggregate shared seed pairs at checkpoint level	Missing
WiSE-FT	Weight-space base/adapted interpolation control	Missing GPU rescoring
KL/replay tuning	Matched-compute preservation baseline	Missing training

Seed pairs in the SFT+SFT arm share adapters. 10 unordered pairs from 5 seeds therefore cannot be treated as 10 independent checkpoint replicates. The locked analysis must aggregate at checkpoint level and preserve the dependency induced by adapter reuse. WiSE-FT and the preservation-tuning arm need compatible retained weights or new training and rescoring.

7.4 Heterogeneity and systems analysis

The heterogeneity hypothesis requires a larger, lineage-diverse checkpoint panel. Base competence will be measured on a locked development cohort; the outcome will be prospective composition-minus-SFT transfer macro-AP on a disjoint cohort. Model identity and lineage must remain visible rather than being hidden behind one correlation coefficient.

Every arm will report end-to-end latency, throughput, peak accelerator memory, training accelerator-hours, and an inference-compute proxy on fixed hardware with a hashed measurement protocol. This separates matched training compute from matched inference compute. Weight-space composition may have a one-pass serving advantage; output composition and SFT+SFT are two-pass systems and must not be described as cost-free.

8 Implementation and reproducibility

The current workspace has two executable boundaries. First, `code/build_pilot_artifacts.py` validates the exact reviewed retrospective JSON and its metadata, then generates all TeX macros and result tables plus a SHA-256 manifest. Its check mode fails when a generated file is missing or stale. Second, `code/validate_protocol.py` validates the prospective contract. Draft mode surfaces the known missing fields; locked mode rejects them.

The current full cloud archive was checksum-verified during the Paper A audit but is not distributed in this repository. The public score cache can reverify the score and public-manifest bindings, but it cannot independently rehash omitted adapter bytes and complete run metadata. Third-party end-to-end re-verification therefore remains incomplete until those artifacts are published at a durable immutable location. This limitation is separate from, and does not excuse, the need for an uninspected prospective Paper B cohort.

9 Limitations and broader scope

The pilot includes 4 compact checkpoints from only two broad model lineages, one guard SFT recipe, English input-prompt classification, benchmark-macro AP, and 7 AP datasets. It does not cover response moderation, tool calls, multi-turn dialogue, multilingual policies, arbitrary new taxonomies, frontier hosted APIs, adversarial production traffic, or human escalation workflows. Benchmark label policies differ, and macro-averaging does not make them interchangeable.

The base is single-pass and lacks base-training-seed replication. The bootstrap is conditional on the selected panel. Transfer rows were not used for SFT, but prior benchmark exposure prevents a prospective interpretation. The current operator was analyzed after Paper A’s lock, and the apparently strongest ablation was visible before this draft. Threshold-free ranking improves while the common-threshold diagnostic misses its calibration FPR target. No standardized latency, throughput, memory, or compute results exist yet. These are study-design constraints, not caveats that can be repaired with stronger prose.

10 Conclusion

In this retrospective fixed panel, a fixed average of separately calibrated base and SFT unsafe scores exchanges lower represented-source AP for higher dataset-held-out transfer AP relative to SFT. The comparison with the base is heterogeneous, and the composition misses the transferred FPR target. The result is strong enough to justify a focused prospective study, but not strong enough to establish dominance, a mechanism, a universal remedy, or a deployment recommendation. The next scientific step is not a more confident headline. It is a separately locked evaluation on a genuinely uninspected cohort with an equal-cost SFT+SFT control, actual WiSE-FT rescoring, matched-compute preservation tuning, independent competence measurement, and complete systems costs.

References

- [1] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- [2] Ananya Kumar, Tengyu Ma, Percy Liang, and Aditi Raghunathan. Calibrated ensembles can mitigate accuracy tradeoffs under distribution shift. In *Proceedings of the Thirty-Eighth Conference on Uncertainty in Artificial Intelligence*, volume 180 of *Proceedings of Machine Learning Research*, pages 1041–1051, 2022. URL <https://proceedings.mlr.press/v180/kumar22a.html>.
- [3] Reza Rahimi. The benchmark chooses the winner: Measuring fine-tuning specialization across safety-guard benchmarks. Companion manuscript and reproducibility artifacts, 2026. Retrospective clean-v2 study.
- [4] Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S. Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig

- Schmidt. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 23965–23998, 2022. URL <https://proceedings.mlr.press/v162/wortsman22a.html>.
- [5] Mitchell Wortsman, Gabriel Ilharco, Jong Wook Kim, Mike Li, Simon Kornblith, Rebecca Roelofs, Raphael Gontijo-Lopes, Hannaneh Hajishirzi, Ali Farhadi, Hongseok Namkoong, and Ludwig Schmidt. Robust fine-tuning of zero-shot models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. URL <https://arxiv.org/abs/2109.01903>.

A Retrospective publication contract

The manuscript-consumed evidence has status `clean_v2_retrospective_estimation`. Its generated manifest records the full composition, Paper A lock, score, and composition-analyzer hashes. The TeX source contains no manually entered pilot AP result. The observed point contrasts are computed from the absolute panel estimates, while bootstrap means are retained under separately named macros. This prevents the common reporting bug of presenting a bootstrap-resample mean as though it were the original observed statistic.

The renderer is intentionally pinned to the reviewed retrospective object. A changed analyzer or new result must be reviewed and explicitly re-anchored; it cannot silently alter the draft. The future prospective pipeline should write into a separate Paper B artifact namespace rather than overwriting this preview.

B Prospective completion checklist

Before the draft can become a claim-bearing paper, the following items must all be complete:

1. select, document, and hash a genuinely uninspected prospective cohort;
2. justify the represented-source noninferiority margin without using prospective outcomes;
3. freeze the operator hierarchy, model panel, estimands, multiplicity policy, RNGs, and family/seed resampling design;
4. implement and verify SFT+SFT, WiSE-FT, and matched-compute KL/replay controls;
5. measure base competence on a disjoint development cohort and expand the checkpoint panel;
6. bind source, configuration, environment, container/runtime, and hardware measurement records in a separate Paper B lock;
7. execute the locked analysis once, preserve failures, and report latency, throughput, peak memory, training compute, and inference compute for every arm; and
8. publish sufficient immutable artifacts for independent verification under applicable data and model licenses.